Jaipur City Transport Plan Report

City: Jaipur

AI-Generated Sustainable Transport Plan

29 June 2026

Executive Summary

The city of Jaipur faces significant transport challenges, including a large demand gap in public transport and high levels of congestion. The proposed plan recommends the development of BRT corridors and a citywide cycling network to address these challenges. However, the plan has some material flaws, including an over-reliance on BRT and a lack of consideration for other modes of transport. The plan aims to increase the public transport share from 19% to 50% by the horizon year, but requires revision to address the identified weaknesses. The revised plan should consider alternative modes of transport, integrate cycling as a feeder mode, and improve the integration with the existing transport network. By addressing these challenges, Jaipur can create a more sustainable and efficient transport system.

Approach

The plan was developed using a multi-agent method, involving data analysis, design with ridership and cost models, and review. This approach allowed for the identification of key transport challenges and the development of targeted solutions. However, the approach also highlighted the need for further consideration of alternative modes of transport and the integration of cycling as a feeder mode. The use of data analysis and modelling enabled the development of a comprehensive plan, but the results require careful review and revision to ensure the plan meets the city's transport needs.

1. Data Foundation (Agent 1)

The plan is grounded in 354 researched facts drawn from 20 sources (Wikipedia, government mobility plans, traffic studies).

Key sources

- Press Release:Press Information Bureau
- Jaipur City Transport Services Overview | PDF | Bus | Land Transport
- Jaipur Low Floor Bus Route 2024 | Jaipur low floor bus routes, Jaipur public transport map, Jaipur travel guide public transport
- Jaipur
- [PDF] COMPREHENSIVE MOBILITY PLAN FOR JAIPUR
- [PDF] Women's safety in Public Transportation: A case study of Jaipur city ...
- Traffic Volume: Chandpole witnesses highest daily traffic volume in Jaipur with 3.3L vehicles: Report | Jaipur News - The Times of India
- City Palace, Jaipur

2. Demand Analysis (Agent 2)

Key metrics

- Population Current: 5.586 million (2024)
- Population Growth Rate: 2.6% per annum (2024)
- Estimated Daily Trips: 3.25 million (derived from home-based travel survey)
- Current Mode Share: two-wheelers: 27%, cars: 17%, city buses: 18%, non-motorised transport: 32%, metro: 0.5%
- Existing Pt Capacity: 1.2-1.5 million person-trips per day (peak-hour bus capacity)
- Pt Demand Gap: shortfall of approximately 0.3-0.6 million person-trips per day (based on designed capacity of 1.8 million trips)

Mobility patterns

mode share: two-wheelers: 27%, cars: 17%, city buses: 18%, non-motorised transport: 32%, metro: 0.5%; peak patterns: morning rush hour: 32 minutes 15 seconds for a 10 km trip, evening rush hour: 36 minutes 9 seconds for a 10 km trip; key od pairs: not specified, but major traffic corridors include Ajmer Road, Apex Circle, Ramganj, Jawahar Nagar, JDA Circle, Khasa Kothi, Chomu Pulia, Shanti Nagar Circle; commute times: average travel time for a 10 km drive: 29 minutes 16 seconds

Current demand

total daily trips: 3.25 million (derived from home-based travel survey); current pt capacity vs demand: shortfall of approximately 0.3-0.6 million person-trips per day (based on designed capacity of 1.8 million trips); underserved areas: 40% of peripheral residential zones lack any transit stop within an 800-metre walking distance

Future demand

growth rate: 2.6% per annum (2024), 2.5% per annum (2031), 2.3% per annum (2041), 1.4% per annum (2051 and 2055); projected population: 66.42 million (2031), 83.15 million (2041), 95.76 million (2051), 101.3 million (2055); projected trips: not specified; trip length: expected to grow from 7.9 km in 2009 to 9.5 km in the 2031 'With Mass Transit System' scenario; motorization trend: two-wheelers and cars together accounted for 83% of Jaipur's registered vehicles in 2017

Capacity gaps

trips that cannot be served: approximately 0.3-0.6 million person-trips per day (based on designed capacity of 1.8 million trips); areas with no pt: 40% of peripheral residential zones lack any transit stop within an 800-metre walking distance; worst times: Friday, 17 October 2025, when congestion peaked at 90% during the 6 pm period

Public-transport deficiencies

coverage: 40% of peripheral residential zones lack any transit stop within an 800-metre walking distance; frequency: medium-frequency routes operate with 15-20 min headways, and low-frequency routes in peripheral areas run every 25-40 minutes; reliability: not specified; comfort: not specified

Bottlenecks

- location: Ajmer Road (Toll Plaza); reason: high traffic volume; severity: 11,207 peak PCU and 116,735 daily PCU, representing 10% of total traffic
- location: Apex Circle; reason: high traffic volume; severity: 2.9 lakh vehicles daily
- location: Ramganj; reason: high traffic volume; severity: 2.8 lakh vehicles daily

3. Proposed Solutions (Agent 3)

Ring Road BRT from Ambabari to Sindhi Camp

Ambabari -> Sindhi Camp | length 4.2 km, 4 stops (length: Nominatim geocode + haversine x1.3 detour)

Mode	Daily Rid.	Peak/hr	Capacity	Capex (cr)	Break-even	Verdict
CYCLING	26,719	3,206	5,000	6.62	Never (subsidy)	VIABLE
BRT	42,750	5,130	12,000	192.6	8.5 yr	MARGINAL
METRO	106,875	12,825	45,000	1402.5	9.2 yr	VIABLE

Recommended: BRT

Why: BRT is the most suitable mode for this corridor as its peak capacity of 12000 pphpd comfortably covers the peak-hour demand of 5130. The capital cost of BRT is significantly lower than metro, making it a more cost-effective option. Cycling is not recommended as the primary mode due to its lower peak capacity and lower daily ridership.

Alternatives rejected: Metro is rejected due to its high capital cost and oversized capacity. Cycling is rejected due to its lower peak capacity and lower daily ridership.

Dhani Govindapura to Ambabari

Dhani Govindapura -> Ambabari | length 10.0 km, 8 stops (length: LLM estimate (geocoding unavailable))

Mode	Daily Rid.	Peak/hr	Capacity	Capex (cr)	Break-even	Verdict
CYCLING	15,000	1,800	5,000	15.75	Never (subsidy)	VIABLE
BRT	24,000	2,880	12,000	388.5	34.1 yr	NOT VIABLE
METRO	60,000	7,200	45,000	2775.0	292.1 yr	NOT VIABLE

Recommended: CYCLING

Why: Cycling is the most suitable mode for this corridor as the peak-hour demand is relatively low and can be covered by the peak capacity of cycling. The capital cost of cycling is also significantly lower than BRT and metro, making it a more cost-effective option.

Alternatives rejected: BRT and metro are rejected due to their high capital costs and oversized capacities.

Ambabari to Kanakpura Railway Station via Ajmeer Road

Ambabari -> Kanakpura Railway Station | length 9.9 km, 10 stops (length: Nominatim geocode + haversine x1.3 detour)

Mode	Daily Rid.	Peak/hr	Capacity	Capex (cr)	Break-even	Verdict
CYCLING	25,250	3,030	5,000	15.59	Never (subsidy)	VIABLE
BRT	40,400	4,848	12,000	399.82	19.6 yr	NOT VIABLE
METRO	101,000	12,120	45,000	2848.75	33.4 yr	NOT VIABLE

Recommended: BRT

Why: BRT is the most suitable mode for this corridor as its peak capacity of 12000 pphpd comfortably covers the peak-hour demand of 4848. The capital cost of BRT is significantly lower than metro, making it a more cost-effective option. Cycling is not recommended as the primary mode due to its lower peak capacity and lower daily ridership.

Alternatives rejected: Metro is rejected due to its high capital cost and oversized capacity. Cycling is rejected due to its lower peak capacity and lower daily ridership.

Citywide recommendations

A citywide cycling network should be developed to provide first-last-mile connectivity to the BRT and metro corridors. The BRT corridors should be prioritized for implementation, followed by the development of a comprehensive cycling network.

Assumptions & caveats

- The route lengths and population densities used in the analysis are estimates and may vary in reality.
- The demand projections are based on current trends and may change over time.
- The capital costs and implementation timelines are estimates and may vary depending on various factors such as land acquisition and construction costs.

4. Independent Review (Agent 4)

Verdict: NEEDS_REVISION

The plan has several strengths, including a thorough analysis of the city's transport problems and a clear presentation of the proposed solutions. However, there are some material flaws that need to be addressed, such as the over-reliance on BRT and the lack of consideration for other modes of transport. Additionally, some of the corridors have peak demands that are not well-matched with the proposed mode's peak capacity.

Key weaknesses identified

- Over-reliance on BRT
- Lack of consideration for cycling as a feeder mode
- Lack of integration with existing transport network

Improvement suggestions

- Consider alternative modes of transport, such as metro or light rail, for corridors with high peak demands
- Integrate cycling as a feeder mode for BRT corridors
- Improve the integration of the proposed corridors with the existing transport network

Conclusion

In conclusion, the Jaipur City Transport Plan requires revision to address the identified weaknesses and ensure the development of a comprehensive and sustainable transport system. The revised plan should prioritize the integration of cycling as a feeder mode, consider alternative modes of transport, and improve the integration with the existing transport network. By taking a holistic approach to transport planning, Jaipur can create a more efficient, sustainable, and equitable transport system that meets the needs of all citizens. The next steps involve revising the plan, engaging with stakeholders, and implementing the revised plan to create a better transport system for Jaipur.